

TLS Antenna System - Detailed Overview

The Tactical Landing System (TLS) provides precision approach guidance similar to an Instrument Landing System (ILS), but with added flexibility in deployment. This document provides an overview of its main components, their functions, connections, and operational ranges.

Component	Function	Connects To	Range (NM)
ASA (Azimuth Sensor Assembly)	Measures azimuth (lateral guidance)	Aircraft, RCU	~0.03
ESA (Elevation Sensor Assembly)	Measures glide slope (elevation)	Aircraft, RCU	0.05–0.1
ATA (Azimuth Transmit Antenna)	Provides azimuth transmission	Aircraft	Local
LOC (Localizer Antenna)	Runway centerline guidance	Aircraft	Up to 25
Glide Slope Antenna	Provides glide path guidance	Aircraft	Up to 18
Uplink Tower	TLS guidance transmission	Aircraft	60–120
SSR (Secondary Surveillance Radar)	Provides radar surveillance	Aircraft, ATC	Up to 100
RCU (Remote Control Unit)	System monitoring & control	ASA, ESA, Cal/BIT	Local
Cal/BIT	Calibration & Built-in Test	RCU	Local

TLS vs ILS

Feature	TLS	ILS
Deployment	Mobile / Flexible	Fixed installation
Coverage	Up to 120 NM (via uplink)	Typically ~18 NM
Guidance	Relies on aircraft transponder code	Signal available to all aircraft
Radar Integration	Integrates with SSR/PAR	Standalone with Localizer/Glide slope

TLS Advantages Over ILS

Advantage	TLS	ILS
Deployment	Mobile and flexible deployment, usable in temporary or permanent locations	Fixed to runway location only
Coverage	Up to 120 NM with uplink tower	Typically ~18 NM
Aircraft Guidance	Guidance based on aircraft-specific transponder code (individualized)	General signal available to all aircraft
Radar Integration	Seamless integration with SSR and PAR systems	Independent system (Localizer/Glide slope only)
Operational Flexibility	Effective in complex terrain and temporary sites	Limited to equipped and fixed runways
Safety	Operates without interfering with nearby runways or airports	Potential interference with neighboring systems

Summary: - TLS provides CAT I precision guidance. - Offers flexibility in deployment compared to fixed ILS systems. - Uses aircraft transponder codes for individual guidance. - Coverage extends up to 120 NM via uplink tower. - Integrates with SSR for surveillance up to 100 NM. - Effective solution for temporary or challenging environments. - Clear operational advantages over ILS in mobility, flexibility, and integration.